What Does a Frozen Video Prior Contribute to Low-Light Enhancement?

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Abstract

A frozen video representation is an appealing source of conditioning for low-light enhancement, but its usefulness does not establish which parts of the representation a restoration network needs. We study a fixed Restormer-style pixel branch conditioned on frozen V-JEPA 2.1 tokens. On held-out SDSD sequences, pretrained conditioning improves a matched zero-token arm by +1.290 dB PSNR-Y and -0.0240 LPIPS after a fixed 20,000*-step schedule. The sequence-paired intervals exclude zero and all three training seeds agree in sign, although effect sizes vary substantially. A corrected single-seed random-encoder control finds lower LPIPS for pretrained tokens, while its PSNR-Y difference remains unresolved. In leave-one-domain-out training, each of three held-out domains improves on at least one of PSNR-Y and LPIPS without significant degradation of the other. Inference interventions show that a trained branch tolerates a substantial reduction of within-clip token variation: projecting centred tokens onto four principal directions while retaining their full clip mean changes PSNR-Y by +0.426 dB. A fixed training-set mean and basis perform worse; this contrast changes both mean and basis. An exploratory validation control finds contributions from the clip mean and, conditional on that mean, clip-specific directions. Spatial and temporal averaging have inconclusive effects rather than demonstrating equivalence. Hybrid keys modestly improve LPIPS. Spatial grounding does not resolve an improvement, and exposure decoupling degrades reconstruction under the tested configurations. Two temporal-loss variants differ in latent directional agreement, while both worsen perceptual quality and a pixel-space temporal diagnostic. These results characterize the benefits and limits of one frozen video prior under a controlled restoration protocol. We release per-sequence measurements and separate corrected controls from those awaiting reconstruction validation; none of the projection experiments establishes that the encoder can be removed.*

1. Introduction

Low-light video enhancement must recover useful appearance from weak and noisy observations. Paired supervision provides a direct learning signal, but the camera, illumination and scene distributions represented in one dataset need not cover those encountered at deployment [9]. A frozen representation learned from other video is therefore a plausible source of additional conditioning. V-JEPA [1, 2], trained by predicting masked spatio-temporal representations, offers such a source. Its predictive training objective motivates asking whether a restoration model benefits from temporal structure as well as appearance.

We investigate that question with a fixed readout: a small Restormer-style pixel branch reads frozen V-JEPA 2.1 tokens through zero-initialised residual cross-attention. The encoder and restoration family remain fixed while we vary conditioning, training domains, token fields and objectives. The purpose is to distinguish an observed gain from claims about the information or computation responsible for it.

Conditioning provides a measurable benefit. After a fixed 20,000-step schedule, pretrained tokens improve a matched zero-token network by $+1.290$ dB PSNR-Y ($[+0.270, +2.449]$) and -0.0240 LPIPS ($[-0.0321, -0.0168]$) on 39 SDSD sequences. All three seeds agree in sign, but the PSNR gains differ considerably in magnitude. In leave-one-domain-out training, the prior improves LPIPS on SDSD and BVI-RLV and PSNR-Y on LOL, with no significant degradation of the other metric. These comparisons establish a benefit for the tested conditioning setup. Reconstructing the random encoder with its training seed corrects a substantial error in the archived attribution control: the single-seed pretrained-minus-random PSNR-Y difference is $+0.604$ dB ($[-0.482, +1.870]$), which is unresolved, while LPIPS improves by -0.0194 ($[-0.0308, -0.0092]$). This supports a conditional perceptual benefit and does not establish a general fidelity advantage over random features.

Tolerance to an intervention is narrower than a compression claim. At inference, retaining four principal directions of each clip's centred token field, together with its full 1024-dimensional mean, changes PSNR-Y by $+0.426$ dB ($[+0.038, +0.868]$). Thus much of the within-clip variation can be removed for these trained readouts. This operation still retains a clip-specific mean, basis and token coordinates; it does not represent the entire clip with four scalars. A fixed training-set mean and rank-four basis cause a substantial loss, but they change two factors at once. An exploratory validation control separates them: the clip mean contributes substantially and, conditional on retaining it, adaptive directions outperform fixed directions (Sec. 5.4; Appendix B). Its rank-four versus full-field comparison remains inconclusive. With its saved inference projection restored, the fixed-rank-64 training control also has an unresolved effect, replacing an earlier apparent penalty. Likewise, the broad intervals for spatial and temporal averaging do not prove equivalence or that those structures are never useful.

More explicit structure need not improve reconstruction. Positional grounding does not resolve a gain, while a tested exposure-decoupled variant degrades the result. Correctly reconstructed hybrid keys improve LPIPS modestly, with unresolved PSNR, so the effect depends on the intervention and metric. A latent temporal objective supplies a sharper example: its cosine form reaches a measured agreement of consecutive-tubelet differences of $+0.089$, yet worsens LPIPS by $+0.0262$ and the tLP gap by $+0.0066$. Engaging a proposed mechanism and improving the restoration task are different empirical requirements.

Protocol and scope. The initial short training budget produced conclusions that changed after longer training, so the main estimates use final checkpoints from the same fixed schedule. Intervals bootstrap paired test sequences after averaging seeds; they are conditional on those trained models rather than estimates of unrestricted training variability. We report corrected random and projection controls separately from archived measurements, and keep unverified architectural evaluations outside the claims. The paper contributes controlled measurements of conditioning benefit and held-out-domain transfer, inference interventions that bound what a trained readout relies on, and a case where better latent temporal agreement accompanies worse restoration. Its conclusions concern this encoder, readout family and evaluation protocol.

2. Related work

Low-light video enhancement. Video restorers combine spatial reconstruction with temporal redundancy through decomposition [17], spatial–temporal compatible learning [22], recurrence [13], state-space models [7], or diffusion [4, 9]. Image methods provide strong frame-wise alternatives, including Retinexformer [3], HVI-CIDNet [18], DarkIR [5], and zero-reference methods [6, 11]. BVI-RLV [9] makes sensor and scene transfer explicit. Our main comparison holds the restoration architecture and training data fixed and changes its conditioning; a separately retrained Retinexformer provides context rather than a claim of universal superiority.

Frozen representations for restoration. Pretrained features can judge reconstruction quality through perceptual losses [8, 21], or condition the restorer directly. DA-CLIP [10] uses adapted vision–language embeddings, and pretrained features also guide restoration networks [16] and diffusion restorers [12]. We study the latter role with a frozen video encoder: V-JEPA [1, 2], learned by predicting masked spatio-temporal representations. Its training objective motivates a temporal prior, but does not establish that a downstream restorer uses temporal information. We measure that distinction by intervening on the tokens and on the training objective.

Representation versus downstream use. A representation may contain information that a particular readout does not use. Our experiments therefore separate encoder sensitivity, restoration quality, and the sensitivity of a trained restorer to token interventions. Likewise, low rank after training is a property of that model under a specified intervention; it does not by itself establish a smaller trainable model, a cheaper encoder, or equivalence between two conditioning distributions.

3. Questions and scope

Let $\mathbf{X} \in [0,1]^{3\times T\times H\times W}$ be a low-light clip and $\mathbf{Y}$ its registered well-lit target. A frozen encoder Φ supplies token fields $Z^{(\ell)} = \Phi^{(\ell)}(\mathbf{X})$ to a trainable pixel branch $\mathcal{P}_\theta$, which predicts $\hat{\mathbf{Y}}$. We ask three questions under a fixed reconstruction protocol.

Does conditioning help? We train the same pixel branch from scratch with either pretrained tokens or zero tokens. The zero-token arm retains the conditioning modules and their parameters. This controls the branch architecture and training schedule, while testing the benefit of providing the pretrained representation. Attributing that benefit specifically to pretraining also requires a correctly reproduced random-encoder arm. We report its corrected single-seed evaluation separately and distinguish perceptual from fidelity evidence (Sec. 5.2).

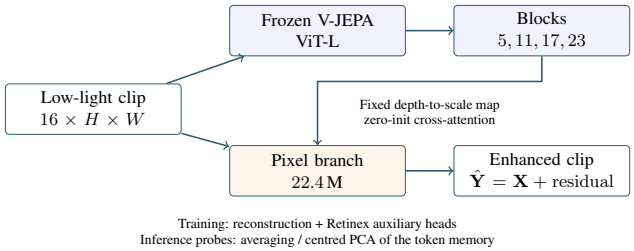


Figure 1. The studied Seer-v2 model. A frozen video encoder supplies four token fields to a small trainable pixel branch through fixed scale assignments. Token interventions change this memory while leaving the trained branch fixed. Auxiliary reflectance and illumination heads are used only during training.

Which parts of the field matter after training? We apply spatial averaging, temporal averaging, and principal-component projection to a model trained with the full field. Each intervention is compared with its own identity control on the same checkpoint and clips. Preserved performance establishes tolerance to that particular replacement. A degradation can reflect lost information or a shift in conditioning, so it cannot alone identify the minimum representation needed for training.

Does temporal information improve the reconstruction? We first test whether the frozen encoder responds to frame order, then test whether the restoration does. Finally, we add a loss that directly matches temporal differences in the frozen feature space. Improving its diagnostic is distinct from improving perceptual quality or pixel-space temporal behaviour. These questions are evaluated at the final checkpoint of a fixed schedule; short-budget experiments are exploratory unless explicitly identified.

4. Seer-v2 and the interventions

4.1. A fixed restoration backbone

The pixel branch is a five-scale Restormer-style U-Net [20] with 22.4 M trainable parameters. Its widths are $(20, 40, 80, 160, 320)$, encoder block counts $(2, 4, 4, 6, 8)$, and decoder block counts $(6, 4, 4, 2)$. Frame-wise blocks perform most of the reconstruction; a residual 3-D convolution and cross-frame attention at the coarsest scale provide temporal mixing. Local illumination cues—luma, channel maximum, local mean, low-pass luma, and gradient magnitude—are projected into each encoder and decoder scale. A zero-initialised residual head gives $\hat{\mathbf{Y}} = \mathbf{X} + \mathcal{P}_\theta(\mathbf{X}; Z)$.

Frozen memory. We use the released V-JEPA 2.1 ViT-L checkpoint distilled from ViT-G [1]. The encoder is frozen throughout restoration training. It receives 16 frames resized to 256×256, producing $8 \times 16 \times 16 = 2048$ tokens of width 1024 at each of blocks $\mathcal{L} = \{5, 11, 17, 23\}$ (zero-indexed). At spatial scales $1/2, 1/4, 1/8, 1/16$, the pixel branch reads these four fields respectively, with the same assignment in the decoder. There is no learned mixture across depths.

For frame t, $Z_{s,t}$ contains the 256 spatial tokens from tubelet $\lfloor t/2 \rfloor$ at the assigned depth. Encoder and decoder features at scale s read this memory through residual multi-head cross-attention,

$$F_{s,t} \leftarrow F_{s,t} + W_o^s \,\mathrm{Attn}(F_{s,t}W_Q^s, Z_{s,t}W_K^s, Z_{s,t}W_V^s). \quad (1)$$

The output projection W_o^s starts at zero. The default attention has no added query–key positional correspondence; spatial grounding is a separate intervention.

Objective. Training-only heads predict reflectance $\hat{R}$ and illumination $\hat{L}$. The default loss is

$$\mathcal{L}_0 = 1.5\,\rho(\hat{\mathbf{Y}}, \mathbf{Y}) + 0.15\,\rho(\hat{R} \odot \hat{L}, \mathbf{Y}) + 0.02\,\mathrm{TV}_{xy}(\hat{L}), \quad (2)$$

where ρ is the mean Charbonnier loss with $\epsilon = 10^{-3}$ and TV_{xy} is spatial total variation. Inference uses the clipped residual prediction without exposure or colour calibration.

4.2. Intervening on the memory

Spatial averaging replaces each tubelet's tokens with their spatial mean; temporal averaging replaces each spatial position with its mean over tubelets. Both broadcast the result to the original grid, preserving the attention tensor shapes.

For one clip and tap, write $Z \in \mathbb{R}^{2048 \times 1024}$ and let μ_Z be its mean token. If $V_{Z,k}$ contains the top k principal directions of $Z - \mathbf{1}\mu_Z$, the per-clip intervention is

$$\mathcal{T}_k^{\mathrm{clip}}(Z) = \mathbf{1}\mu_Z + (Z - \mathbf{1}\mu_Z)V_{Z,k}V_{Z,k}^\top. \quad (3)$$

This preserves a full 1024-dimensional clip mean; only the centred token variation is restricted to rank k. The fixed intervention uses a mean μ_{tr} and directions $V_{\mathrm{tr},k}$ estimated separately for each tap from training clips,

$$\mathcal{T}_k^{\mathrm{fixed}}(Z) = \mathbf{1}\mu_{\mathrm{tr}} + (Z - \mathbf{1}\mu_{\mathrm{tr}})V_{\mathrm{tr},k}V_{\mathrm{tr},k}^\top. \quad (4)$$

The comparison changes both the mean and the basis; it does not isolate basis adaptivity alone. An exploratory validation control crosses the two means and the two bases using the same reconstruction formula, alongside identity and broadcast clip-mean controls (App. B). The adaptive directions always come from clip-centred tokens, including when the reconstruction uses the training-set mean.

4.3. Testing a temporal objective

Let $\Delta Z_t = Z_{t+1} - Z_t$ be the block-23 token difference along tubelets. We add $0.3\mathcal{L}_\Delta$ to Eq. (2), with gradients through the frozen encoder to the predicted pixels. The relative form

is $\mathcal{L}_\Delta^{\text{rel}} = \|\Delta\Phi(\hat{\mathbf{Y}}) - \Delta\Phi(\mathbf{Y})\|_1 / \max(\|\Delta\Phi(\mathbf{Y})\|_1, \epsilon)$. A temporally constant prediction scores approximately one, so shrinking the variation can reduce this objective without reconstructing the dynamics. The cosine form instead uses

$$\mathcal{L}_\Delta^{\cos} = 1 - \overline{\cos(\Delta\Phi(\hat{\mathbf{Y}}), \Delta\Phi(\mathbf{Y}))} + \left| \log \frac{\max(\|\Delta\Phi(\hat{\mathbf{Y}})\|_1, \epsilon)}{\max(\|\Delta\Phi(\mathbf{Y})\|_1, \epsilon)} \right|, \tag{5}$$

where cosine is computed per token over channels and the ℓ_1 quantities are mean absolute magnitudes. We record both directional agreement and the magnitude ratio to check what the loss actually changes.

4.4. Training

All reported converged arms use a fixed 20,000-step schedule. The pixel branch starts from scratch with AdamW, learning rate 2×10^{-5}, weight decay 10^{-4}, betas $(0.9, 0.99)$, batch one and eight-step gradient accumulation. Training uses random 256×256 crops of 16-frame clips, flips, gradient clipping at one, bf16 autocast, and gradient checkpointing. Validation is monitored without selecting a checkpoint; the final checkpoint is scored. The default objective contains no latent temporal term.

5. Experiments

5.1. Protocol

Data and evaluation. In-domain training uses the fixed SDSD [14] split: 136 training sequences, 16 validation sequences, and 39 test sequences. Scene groups are kept together when defining validation; sequences identified as leaking between training and test were removed before this round. For transfer we treat SDSD, BVI-RLV [9], and LOL [15, 19] as three domains. BVI-RLV has 48/16/16 train/validation/test clips from 24/8/8 scenes. The LOL domain pools LOL-v1, LOL-v2-real, and LOL-v2-synthetic, with 215 test images in total. Leave-one-domain-out training samples each of the two source domains with probability 1/2; neither source arm sees the held-out domain.

Videos use the centre 16 frames and a 256×256 centre crop. Frames are paired by sorted index rather than filename. LOL images are evaluated at full resolution, repeated to form the encoder's clip input; the single image prediction is scored. PSNR-Y is computed after uint8 rounding, using the evaluator's BT.601 luma conversion. LPIPS uses AlexNet [21]. Temporal diagnostics include LPIPS between adjacent frames (tLP), reported as the absolute gap to the target clip's value after seed-averaging within each arm. Such diagnostics remain sensitive to motion and are not interchangeable with perceptual reconstruction quality.

Conditioning / model	PSNR-Y ↑	LPIPS ↓
Zero tokens	23.300	0.2400
Pretrained tokens	24.589	0.2160
Retinexformer, retrained	22.621	0.2115

Table 1. SDSD test, $n = 39$. Seer-v2 arms share architecture, from-scratch recipe and the three nominal seeds. They do not share initial pixel weights or training random draws (App. A). Retinexformer uses the same training split but its own architecture and training schedule.

Decision rule. The main in-domain comparison has three seeds; transfer and most interventions have two. Differences are paired by sequence after averaging within each arm over seeds. We use 10,000 sequence-bootstrap samples for 95% intervals. A claimed effect must exclude zero and have the same sign in every seed. These intervals describe test-sequence variation conditional on the trained models; the sign check does not replace estimating uncertainty over a larger population of training seeds. Unless specified otherwise, Δ means intervention minus reference, so higher PSNR and lower LPIPS are better.

5.2. The prior helps in domain

Pretrained conditioning improves the matched zero-token arm by $+1.290$ dB ($[+0.270, +2.449]$) and -0.0240 LPIPS ($[-0.0321, -0.0168]$). All three seeds agree in sign on both metrics. The PSNR effect varies substantially across seeds, so its mean is evidence of benefit under this protocol rather than a promise of a fixed improvement on every run. Because both arms start from scratch, this comparison avoids inheriting a branch already adapted to the pretrained tokens. The matched comparison alone does not separate the value of pretraining from every possible alternative feature source. The archived random control instantiated a different random encoder at evaluation. Reusing the training seed instead yields 24.084 dB and 0.2366 LPIPS for that seed-zero checkpoint. Against the corresponding pretrained seed, the differences are $+0.604$ dB ($[-0.482, +1.870]$) and -0.0194 LPIPS ($[-0.0308, -0.0092]$). Only LPIPS resolves a benefit. This is a single trained random-control seed, so its sequence interval cannot establish consistency over training seeds.

Comparison with a retrained restorer. Retinexformer [3] is retrained for 160,000 steps on the same SDSD training split. Seer-v2's mean PSNR margin is $+1.968$ dB ($[-0.365, +4.543]$), but its median is -0.307 dB and it wins only 19/39 sequences. Its LPIPS difference is $+0.0045$ ($[-0.0287, +0.0363]$). Neither interval excludes zero. The uneven sequence-level margin and different model sizes preclude a claim of uniform superiority. Our split and evaluation

crop also differ from published benchmark tables, so those absolute scores are not directly comparable.

5.3. The prior transfers under held-out-domain training

The registered transfer criterion requires improvement on at least one metric in at least two held-out domains, without significant degradation of the other metric. All three domains satisfy it (Table 2). The metric that improves differs: perceptual distance on SDSD and BVI-RLV, and luminance fidelity on LOL. This supports transfer of useful conditioning under the specified source mixtures. It does not establish simultaneous improvement on every metric or universal transfer to arbitrary sensors. LOL is an image-domain test, so its improvement supplies no evidence of temporal transfer.

5.4. What a converged branch tolerates

Keeping four principal directions of each clip's centred tokens improves PSNR-Y by $+0.426$ dB ($[+0.038, +0.868]$), with all three seeds agreeing. The corresponding 1024-dimensional mean and the basis itself remain clip-dependent. This is evidence that much of the within-clip token variation can be removed after training; it is not a four-dimensional representation of the whole clip.

Spatial or temporal averaging does not produce a significant PSNR change. Their intervals are broad: the data still admit losses of roughly 0.68 dB and 0.18 dB, respectively. Absence of a detected degradation is therefore weaker than equivalence or proof that the original field carries no useful spatial or temporal detail.

Fixed rank-4 and rank-16 replacements significantly degrade PSNR; fixed rank-64 does not. The contrast establishes that this fixed low-rank replacement is less compatible with the learned branch than the per-clip replacement. Because Eqs. (3) and (4) differ in both mean and basis, that contrast cannot assign the gap solely to basis adaptivity. Nor does it establish that the frozen encoder is impossible to replace.

Separating mean and basis on validation. An exploratory follow-up crosses both factors on all 16 SDSD validation sequences (8 scenes, 2 existing seeds; App. B). Holding the clip mean fixed, adaptive rather than fixed rank-four directions change PSNR-Y by $+1.959$ dB, with unadjusted scene-bootstrap interval $[+0.354, +3.799]$ and positive effects in both seeds. Holding either basis fixed also favours the clip mean over the training mean. This implicates both factors on validation; it does not establish a training bottleneck or a cheaper encoder. Rank-four projection versus identity remains inconclusive on this split.

Training-time projection addresses a different question. The archived fixed-rank training evaluation did not restore its input projection. Corrected evaluation applies the saved training-set mean and rank-64 basis at inference as well. Against full-token-trained seeds zero and one, its PSNR difference is -0.061 dB ($[-0.571, +0.457]$), and its LPIPS difference is -0.0002 ($[-0.0041, +0.0040]$). The per-seed PSNR differences have opposite signs. Neither interval resolves a difference, so the earlier significant training-time penalty is withdrawn. These intervals also do not establish equivalence or a faster model: the projection still follows the full encoder.

5.5. Injection changes have mixed outcomes

Restoring the saved hybrid key/value configuration also changes its archived evaluation. Against the corresponding per-tubelet seeds zero and one, hybrid keys change PSNR-Y by $+0.049$ dB ($[-0.425, +0.492]$) and LPIPS by -0.0048 ($[-0.0081, -0.0016]$). LPIPS improves in both seeds; PSNR remains unresolved. This supports a modest perceptual benefit for this tested key/value variant.

Spatial grounding adds positional correspondence between pixel queries and token keys. Over two seeds, its PSNR difference is $+0.029$ dB ($[-0.386, +0.367]$), and its LPIPS difference is -0.0002 ($[-0.0043, +0.0041]$). Neither metric resolves an improvement at convergence.

Exposure decoupling adds a predicted global gain and a supervised exposure objective. In its two-seed comparison, PSNR changes by -1.614 dB ($[-2.312, -0.924]$), while LPIPS changes by $+0.0079$ ($[+0.0034, +0.0124]$), both worse. The spatial and exposure interventions do not justify their added structure under this protocol. A corrected all-depth comparison is limited to one variant seed with different training hardware (App. C); deepest-only taps and learned gates still await verified evaluation.

5.6. A temporal objective engages a different target

The encoder is sensitive to temporal structure on this data. The archived eight-clip probe detects a substantial block-23 token change after shuffling under its relative token-distance measure. This is a measurement of the encoder, separate from the restorer.

The relative temporal loss reduces the latent magnitude ratio to 0.135 and leaves directional agreement at $+0.014$. Its LPIPS and tLP-gap changes are $+0.1826$ ($[+0.1659, +0.1985]$) and $+0.3555$ ($[+0.3391, +0.3724]$), respectively: the loss rewards a smaller latent variation without improving the measured reconstruction.

The cosine form restores the magnitude ratio to 1.001 and reaches directional agreement $+0.089$. Nevertheless LPIPS worsens by $+0.0262$ ($[+0.0128, +0.0412]$) and the tLP gap worsens by $+0.0066$ ($[+0.0044, +0.0088]$). PSNR is unresolved: $+0.049$ dB ($[-0.408, +0.558]$). The archived order probe for this model changes aggregate PSNR by only 0.034 dB after shuffling. These observations distinguish matching latent temporal differences from improving the pixel-space result; the former does not guarantee the latter.

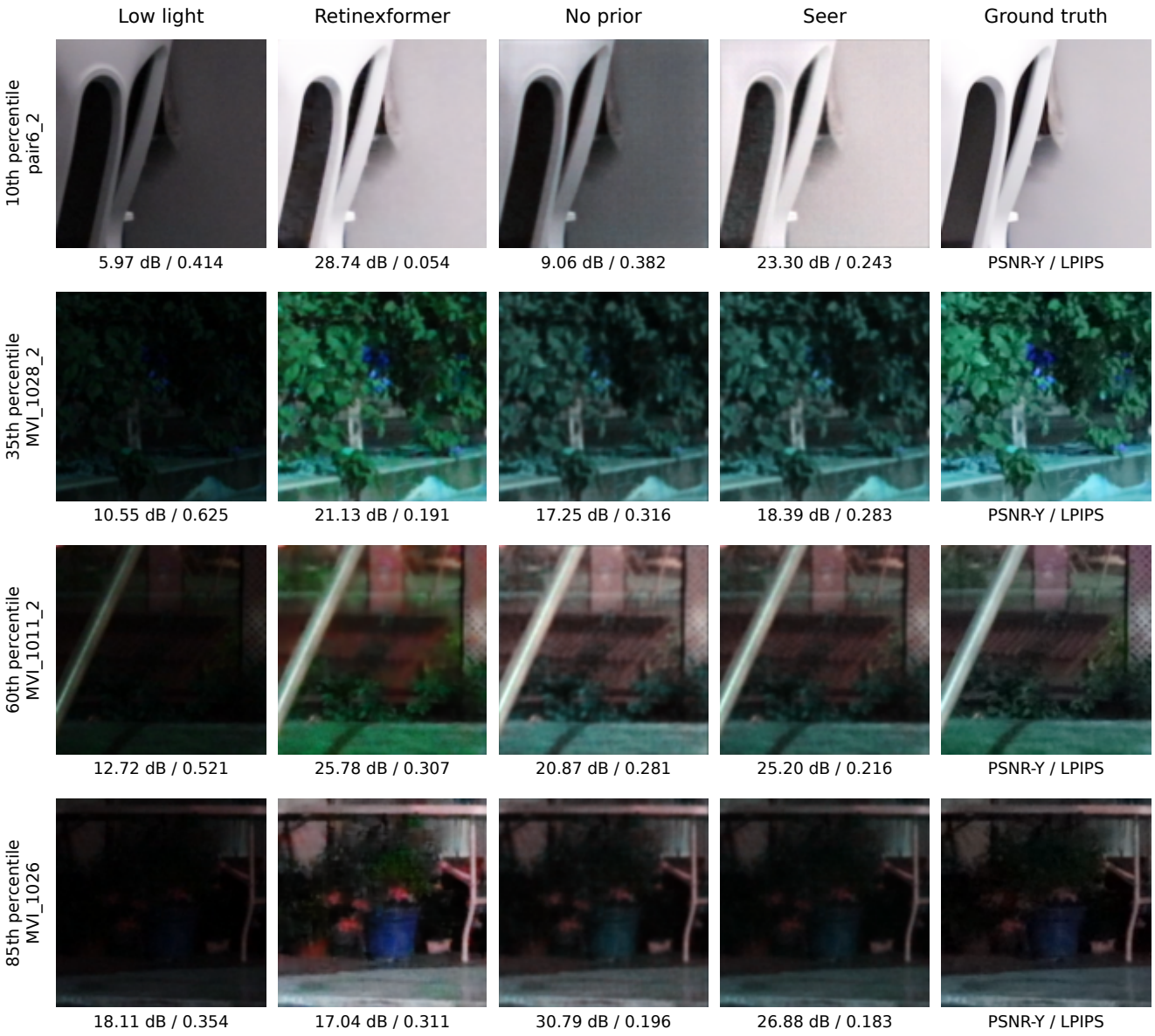


Figure 2. Qualitative comparison, with columns showing input, retrained Retinexformer, zero-token Seer-v2, pretrained Seer-v2, and ground truth. Sequences are selected at four fixed quantiles of the sequence-level PSNR-Y margin between the three-seed pretrained average and Retinexformer. Displayed panel metrics score one middle frame, whereas the tables score the full evaluation clip. Panels use pretrained and zero-token seed zero; checkpoint, source and image hashes accompany the released figure.

Held out	n	Δ PSNR-Y [95% CI]	Δ LPIPS [95% CI]
SDSD	39	+0.292 [-1.875, +2.519]	-0.0257 [-0.0471, -0.0059]
BVI-RLV	16	+0.298 [-1.593, +1.965]	-0.0229 [-0.0435, -0.0017]
LOL	215	+0.881 [+0.390, +1.395]	+0.0011 [-0.0031, +0.0054]

Table 2. Leave-one-domain-out transfer: pretrained minus zero-token conditioning, two seeds per arm. A domain passes the registered rule when one metric improves significantly, every seed agrees on that metric, and the other metric has no significant degradation. SDSD and BVI-RLV pass on LPIPS; LOL passes on PSNR-Y.

Inference intervention	Seeds	Δ PSNR-Y [95% CI]
Per-clip centred rank 4	3	+0.426 [+0.038, +0.868]
Spatial mean	3	+0.336 [-0.685, +1.331]
Temporal mean	3	+0.283 [-0.183, +0.758]
Fixed rank 4	2	-4.275 [-6.835, -1.888]
Fixed rank 16	2	-3.115 [-5.279, -1.142]
Fixed rank 64	2	-0.215 [-1.395, +0.904]

Table 3. Inference-only interventions on full-token-trained checkpoints, paired with their identity controls on the same 39 SDSD clips. Per-clip projection retains the full clip mean; fixed projection uses the training-set mean and basis. The two blocks have different seed counts.

5.7. Why training budget belongs in the conclusion

The initial short-budget round and the converged round lead to different decisions about the conditioning benefit, transfer, and temporal interventions. Consequently, all effect estimates used above come from the fixed 20,000-step schedule, and early results are kept as exploratory records. A conclusion about a small architectural effect requires that the reference and intervention have both been trained long enough to make that comparison meaningful. This is a protocol requirement demonstrated by this study, rather than a proposed universal iteration count.

6. Conclusion

Frozen V-JEPA conditioning benefits a matched low-light restorer in domain and under the specified leave-one-domain-out protocol. The stronger question is what the trained branch uses. It tolerates removal of most centred token variation when its clip mean and a few clip-specific principal directions are preserved; the tested fixed replacement is much less successful. An exploratory validation control separates contributions of the clip mean and, conditional on that mean, adaptive directions; its rank-four versus full-field comparison remains inconclusive. These results characterise the trained readout and do not yet provide a smaller encoder or a validated computational compression recipe. The corrected fixed-rank training control has an unresolved effect rather than the previously reported penalty, while the single-seed random control resolves a perceptual benefit without resolving a fidelity benefit. Hybrid keys modestly improve LPIPS, spatial grounding does not resolve an improvement, exposure decoupling degrades the result, and directly matching latent temporal changes worsens perceptual quality and the measured tLP gap. The study supports measuring downstream use separately from what a pretrained model can represent, and making those measurements on converged, reproducible comparisons.

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Reproducibility statement

The reported result macros are generated by `tools/make_macros_cvpr.py` and `tools/make_audit_macros.py` from archived and separately corrected per-sequence files. The archive includes training and evaluation code, frozen split lists, registered decision rules and available measurements. Section 4 and App. A describe the current model and protocol; App. A describes split membership. Appendix C identifies missing evidence and evaluations withheld pending reconstruction checks. Historical files are preserved and are not silently substituted for verified current results.

Ethics statement

The work uses public research datasets (SDSD, BVI-RLV, LOL) and released model weights under their licences. Low-light enhancement can be used in surveillance; we do not target identification tasks.

A. Implementation and data details

Encoder. The frozen checkpoint is V-JEPA 2.1 ViT-L distilled from ViT-G at 384 pixels. Its input is bilinearly resized to 256×256 and normalised using ImageNet statistics. Tubelet size is two, patch size is 16, and the 24-block encoder is read at zero-indexed blocks $5, 11, 17, 23$ after the corresponding hierarchical normalisation. This produces four 2048×1024 fields. The frozen encoder contributes approximately 304.7 M parameters in addition to the 22.4 M trainable pixel branch. Parameter count of the pixel branch alone is not an inference-cost comparison.

Pixel branch. Attention heads are $(2, 4, 4, 8, 8)$ and the feed-forward expansion is 2.66. The decoder mirrors the four non-bottleneck encoder scales, with skip concatenation and a 1×1 reduction. Pixel-unshuffle and pixel-shuffle operations change resolution. The bottleneck has a residual temporal convolution followed by cross-frame attention; other restoration blocks operate frame-wise. Each stage applies restoration blocks, token cross-attention where enabled, and local illumination cue injection. Reflectance and illumination heads are convolution–SiLU–convolution–sigmoid stacks with three and one output channels, respectively. Their outputs contribute only to the training loss. The residual head is a zero-initialised 3×3 convolution.

Initialisation and optimisation. The converged CVPR comparisons use from-scratch pixel branches, not the warm-started stages in the earlier project snapshot. The frozen representation is the only pretrained part of the default model. The zero-token arm keeps the same cross-attention modules. The arms use the same nominal seed set, but constructing the pretrained encoder consumes random draws before the pixel branch is initialised, whereas the zero-token arm skips that construction. Their initial pixel weights and training random draws are therefore not shared. The comparison controls architecture and optimisation recipe; pairing refers to test sequences, not to identical training randomness. All arms use the optimiser and crop recipe in Sec. 4.4 and score the final checkpoint without validation-based selection. The converged run collection includes both Ascend 910B3 and NVIDIA A100 runs; experiment identifiers, seeds and hardware are listed in the released manifest. The third in-domain seed was completed on different hardware, so the seed collection does not separately estimate hardware effects.

Split provenance. SDSD-indoor contributes $46/6/14$ train/validation/test sequences and SDSD-outdoor $90/10/25$. Validation groups repeated passes of a scene together. BVI-RLV contains both frame rates of each scene; its two clips per scene stay in the same data split. LOL-v1 has $436/49/15$ training/validation/test images, LOL-v2-real $620/69/100$, and LOL-v2-synthetic $810/90/100$. The frozen split files, rather than a split reconstructed at evaluation time, define membership. Retinexformer is retrained on these SDSD training sequences. Results from released checkpoints trained on other splits are excluded from the current main comparison.

Evaluation details. Inputs and targets are paired by sorted frame index, which handles SDSD sequences with filename offsets. The central video clip and spatial crop are fixed before inference. Short clips and images are repeated to 16 frames; images produce a single scored frame. Image sizes are padded to the branch's spatial divisor and cropped back after inference. Raw predictions are clipped to $[0, 1]$ and rounded to uint8 for PSNR. GT-mean rescaling and previous colour-calibration procedures are not used in the main claims. The registered bootstrap unit is the evaluated sequence or image. Paired SDSD passes and BVI-RLV frame-rate clips can share a scene. A post-hoc sensitivity analysis therefore resamples whole scene groups, retaining sequence weights through a total-sum/total-count estimator. The in-domain PSNR-Y and LPIPS benefits and the two video-domain transfer LPIPS benefits retain their direction and zero-excluding intervals. The SDSD transfer LPIPS bound is close to zero. This robustness analysis does not replace more training seeds or eliminate other dependence.

B. Intervention details

Principal components. Per-clip projection centres all 2048 tokens at each tap, obtains the right singular vectors of the centred matrix, and reconstructs it after retaining k directions. The full clip mean is added back. It therefore

Validation condition	PSNR-Y ↑	LPIPS ↓
Identity	24.577	0.2414
Clip mean only	23.283	0.2518
Clip mean, clip basis	24.698	0.2409
Clip mean, fixed basis	22.740	0.2593
Train mean, clip basis	17.735	0.3637
Train mean, fixed basis	17.946	0.3517

Table 4. Exploratory mean–basis diagnostic on 16 validation sequences from 8 scenes, averaged over 2 existing checkpoints. The four crossed conditions use rank-four directions; clip-mean-only broadcasts the mean to every token. These are validation measurements, separate from Table 3's test results.

retains substantially more than k scalar values: the mean, directions, and token-dependent coordinates all remain. The fixed basis uses the first 60 training-split clips in the saved order, accumulating means and second moments separately at each tap; eigendecomposition gives a rank-64 basis, truncated to the requested rank. No target or test label is used in either projection. The per-clip projection nevertheless depends on the test input, as intended, whereas the fixed basis and mean do not.

Exploratory mean–basis control (validation only). The historical test intervention changes both factors. To separate them, we use the existing full-token-trained E17 seeds zero and one on every frozen SDSD validation sequence, without retraining or checkpoint selection. The diagnostic was chosen after the earlier results were inspected; its six conditions and validation-only protocol were recorded in Amendment 21 before its outcomes. It is exploratory rather than a new confirmatory test. All principal-component projections here have rank four. The adaptive directions are computed from clip-centred tokens independently of which mean is used in reconstruction.

We pair differences by sequence after seed averaging and report unadjusted 95% scene-bootstrap intervals, with 10,000 resamples from eight scenes; the archive also reports sequence-bootstrap intervals and every seed effect. At fixed clip mean, replacing the training basis with adaptive directions changes PSNR-Y by $+1.959$ dB ($[+0.354, +3.799]$). At fixed adaptive basis, replacing the training mean with the clip mean changes it by $+6.964$ dB ($[+1.949, +12.803]$); with the fixed basis, the corresponding change is $+4.794$ dB ($[+1.209, +8.924]$). Every seed agrees in sign on these contrasts. The basis contrast at the training mean remains unresolved on PSNR, so the adaptive-basis effect is conditional on the retained mean.

Rank-four adaptive projection also improves on clip-mean-only by $+1.415$ dB ($[+0.351, +2.674]$), with both seeds positive. Against identity, however, the change is $+0.121$ dB ($[-0.089, +0.313]$), and LPIPS effects have opposite signs across seeds. This does not establish equivalence or a validation improvement over the full field. The matched contrasts support contributions from both the input-dependent mean and basis under these interventions. They do not show that either can be obtained cheaply, that a trained model needs only four dimensions, or that the same representation suffices during training.

Interpretation of averaging. Spatial mean retains a separate vector for each tubelet; temporal mean retains a separate vector for each spatial position. Neither transformation removes the frozen encoder's computation. An interval that crosses zero bounds the detected effect but does not establish statistical equivalence without a predeclared equivalence margin.

Temporal measurements. The latent diagnostic reports mean cosine agreement of consecutive-tubelet differences and their mean absolute magnitude ratio. The relative and cosine losses are trained on two seeds each. The cosine form's log-ratio penalty penalises both expansion and contraction. For each sequence and arm, tLP and target tLP are averaged over the arm's seeds, then their absolute difference is taken. Differences between these absolute gaps are paired over sequences. Each individual seed's absolute-gap effect is also checked for the same sign. The order of averaging and taking the absolute value matches the released aggregation tool. This gap uses identical frames and crops and is not a motion-compensated error.

C. Corrected controls and outstanding validation

The archive contains evaluations beyond those used in the current claims. The random-encoder control has now been rescored with CPU initialisation seed 1234, matching the training implementation. Its recorded initial-parameter checksum agrees up to the float-versus-double accumulation difference logged by the NPU runtime; this is an approximate fingerprint, not a full model hash. The corrected JSON records reconstruction settings separately from the historical result. Section 5.2 uses the corrected single-seed comparison and does not retain the earlier large fidelity-attribution claim.

Corrected hybrid cross-frame keys are reported in Sec. 5.5. Restoring all-depth keys for the available seed-two checkpoint gives 25.139 dB, replacing the archived, incorrectly configured 21.568 dB result. Against the corresponding full-token default seed, its differences are $+0.476$ dB ($[-0.276, +1.245]$) and -0.0045 LPIPS ($[-0.0094, +0.0001]$). Both intervals include zero. The variant was trained on Ascend and the reference on A100, with only one available variant seed; this conditional comparison cannot establish a general architectural effect. Its

temporal diagnostics also show tradeoffs, detailed in the audit record. Deepest-only taps and learned gates still require the actual trained configuration and weights before their archived differences can support architectural conclusions. The fixed-rank-64 training control has been reevaluated with its saved projection. Its current comparison uses the same two baseline seed numbers; the old three-baseline-seed comparison is preserved in the audit record. Neither comparison resolves a training-time penalty after correction. It still does not establish a computational compression result.

H20/E28, training from scratch with per-clip centred rank-four tokens, remains unfinished: no verified final evaluation is available. It is neither a positive nor a negative result. The proposed small-network distillation experiment was cancelled by its registered screening rule and was not run; its cancellation supplies no evidence that another encoder or distillation procedure cannot work. Some baseline temporal diagnostics were not included in the original archive. Claims requiring those missing measurements are withheld until they are reproduced under the same protocol. The qualitative panels have been regenerated with checkpoint, source and output hashes.

D. Scope of the conclusions

The main conclusions concern one frozen encoder, one restoration family, and the specified data splits. Few seeds limit inference about training variability, and sequence bootstraps do not replace additional independent runs. SDSD includes low-motion material, while the LOL test contains still images. A video encoder's temporal sensitivity and a readout's weak order sensitivity therefore do not prove that temporal representations are irrelevant to other video-restoration tasks. The tested low-rank replacements describe the behaviour of existing checkpoints. The exploratory validation control separates mean and basis effects under those replacements; future work must separately test efficient encoders and training with the restricted representation.